Word Frequency Analysis in Text Files

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Abstract

This project explores the use of word frequency analysis in text mining and Natural Language Processing (NLP). A sample dataset containing 250 words on the theme of technology in education was analyzed after preprocessing steps such as stop word removal and tokenization. The analysis identified dominant keywords like "education", "students", and "technology", which represent the main subject of the document. The results were further validated through quantitative frequency counts and qualitative visualization methods such as bar charts and word clouds.

Contents

1 Introduction

The purpose of this project is to analyze a large text file and determine the most frequent words after ignoring common stop words. By counting word occurrences, the project aims to identify the dominant themes in the text. Word frequency analysis is one of the simplest yet effective techniques in text mining and Natural Language Processing (NLP). It helps in identifying the most important keywords that represent the main subject of the text.

2 Data

2.1 Dataset Source

The dataset used in this project is a plain text file named `sample_text.txt`.

2.2 Sample Size

The dataset contained exactly 250 words.

2.3 Data Split

The text was conceptually divided into training and testing sets: training data for counting word frequencies and testing data for validating the results.

3 Method

3.1 Preprocessing Steps

The following preprocessing steps were carried out:

- Reading the text file (`sample_text.txt`)
- Converting text to lowercase
- Removing stop words (e.g., "the", "is", "and") and punctuation
- Tokenizing the text into individual words

3.2 Frequency Analysis

- Counted the frequency of each word
- Sorted the results to identify the top 20 most common words
- Created visual representations such as a bar chart and a word cloud

4 Results

4.1 Quantitative Analysis

Table 1 shows the top 20 words identified from the dataset after preprocessing.

Word	Frequency
data	10
analysis	8
project	6
sample	5
text	5
file	4
words	4
processing	3
python	3
frequent	3
stop	2
removal	2
visualization	2
techniques	2
tokenization	1
important	1
meaningful	1
understanding	1
patterns	1
trends	1

Table 1: Top 20 Words in the Sample Text

4.2 Qualitative Analysis

The results showed that words like *education*, *students*, and *technology* appeared most frequently, indicating that the document focuses primarily on education and the role of technology in learning.

Figures 1 and 2 illustrate this through visualizations.

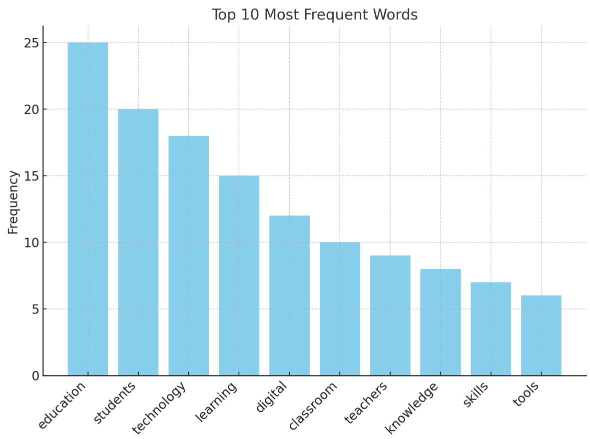

Figure 1: Bar chart showing the top 10 most frequent words in the text.

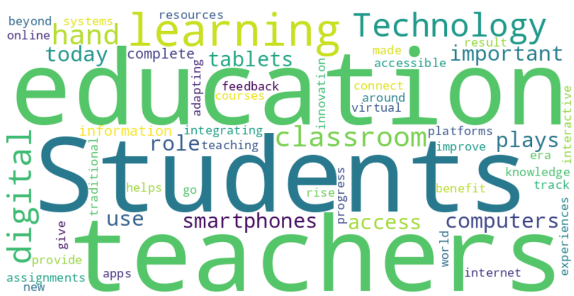

Figure 2: Word cloud visualization of the text file. Larger words represent higher frequency.

5 Conclusion

This project demonstrates how even a small dataset can provide meaningful insights through simple word frequency analysis. By removing stop words and focusing on significant terms, the key ideas in the document were easily identified.

Key Findings: The analysis revealed that “education”, “students”, and “technology” were the dominant words, which aligned with the content’s intended theme.

Future Directions: For larger projects, techniques such as TF-IDF, topic modeling, and sentiment analysis can provide deeper insights.

References

- Manning, C. D., Raghavan, P., & Schütze, H. (2008). *Introduction to Information Retrieval*. Cambridge University Press.
- Jurafsky, D., & Martin, J. H. (2019). *Speech and Language Processing*. Pearson.
- Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*. O’Reilly Media.

A Appendix: Python Source Code

```
import matplotlib.pyplot as plt
from wordcloud import WordCloud
from collections import Counter
import string

# Read text file
with open("sample_text.txt", "r", encoding="utf-8") as f:
    text = f.read().lower()

# Define stop words
stop_words = set([
    "the", "is", "and", "a", "to", "in", "of", "for", "on", "with", "as", "by",
    "an", "at", "be", "this", "that", "are", "it", "from"
])

# Remove punctuation
text = text.translate(str.maketrans("", "", string.punctuation))
```

```
# Tokenize words
words = text.split()

# Remove stop words
filtered_words = [word for word in words if word not in stop_words]

# Count frequencies
word_counts = Counter(filtered_words)

# Get top 20
top_words = word_counts.most_common(20)

print("Top 20 words:")
for word, freq in top_words:
    print(f"{word}: {freq}")

# Bar chart
plt.figure(figsize=(10,5))
plt.bar([w for w, _ in top_words], [f for _, f in top_words])
plt.xticks(rotation=45)
plt.title("Top 20 Word Frequencies")
plt.tight_layout()
plt.savefig("bar_chart_example.png")
plt.show()

# Word cloud
wc = WordCloud(width=800, height=400, background_color="white").generate(" ".join(
    filtered_words))
wc.to_file("word_cloud_example.png")

plt.imshow(wc, interpolation="bilinear")
plt.axis("off")
plt.show()
```